

Evan Edelstein
 EN.605.645.82.SP26
 Module 8 – self check

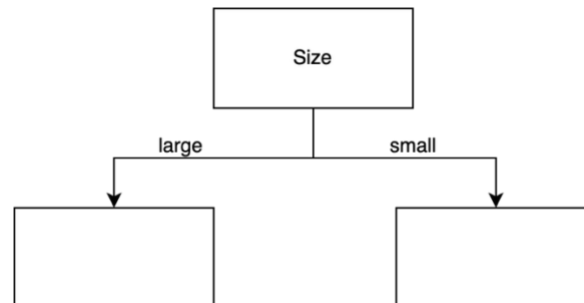
We start with the feature set {Shape, Size, Color}. To pick which feature to partition we can calculate the entropy for each feature. The data size is 15. We start by getting the entropy for each attribute/feature.

Feature	Attribute	Label	Label size	P	Entropy Formula	Entropy Value	Feature Entropy
Shape	round	yes	4	4/7	$-\frac{4}{7} * \log_2(4/7)$	0.461	$7/15 * 0.461 = 0.215$
Shape	round	no	3	3/7	$-\frac{3}{7} * \log_2(3/7)$	0.524	$7/15 * 0.524 = 0.244$
Shape	square	yes	3	3/8	$-\frac{3}{8} * \log_2(3/8)$	0.531	$8/15 * 0.531 = 0.283$
Shape	square	no	5	5/8	$-\frac{5}{8} * \log_2(5/8)$	0.424	$8/15 * 0.424 = 0.226$
Shape							0.969
Size	large	yes	6	6/8	$-\frac{6}{8} * \log_2(6/8)$	0.311	$8/15 * 0.311 = 0.166$
Size	large	no	2	2/8	$-\frac{2}{8} * \log_2(2/8)$	0.500	$8/15 * 0.500 = 0.267$
Size	small	yes	1	1/7	$-\frac{1}{7} * \log_2(1/7)$	0.401	$7/15 * 0.401 = 0.187$
Size	small	no	6	6/7	$-\frac{6}{7} * \log_2(6/7)$	0.191	$7/15 * 0.191 = 0.089$
Size							0.709
Color	blue	yes	0	0/3	0	0.000	$3/15 * 0.000 = 0.000$
Color	blue	no	3	3/3	$-\frac{3}{3} * \log_2(3/3)$	0.000	$3/15 * 0.000 = 0.000$
Color	green	yes	4	4/6	$-\frac{4}{6} * \log_2(4/6)$	0.390	$6/15 * 0.390 = 0.156$
Color	green	no	2	2/6	$-\frac{2}{6} * \log_2(2/6)$	0.528	$6/15 * 0.528 = 0.211$
Color	red	yes	3	3/6	$-\frac{3}{6} * \log_2(3/6)$	0.500	$6/15 * 0.500 = 0.200$
Color	red	no	3	3/6	$-\frac{3}{6} * \log_2(3/6)$	0.500	$6/15 * 0.500 = 0.200$
Color							0.767

We see that Size has the lowest entropy, so we recurse while removing all rows where Size is large and all removing all rows with Size == small.

Shape	Size	Color	Safe?
round	large	blue	no
square	large	green	yes
square	small	red	no
round	large	red	yes
square	small	blue	no
round	small	blue	no
round	small	red	yes
square	small	green	no
round	large	green	yes
square	large	green	yes
square	large	red	no
square	large	green	yes
round	large	red	yes
square	small	red	no
round	small	green	no

This branching also forms the root node in the tree:



Looking at the left branch, we subset the data by removing small size rows.

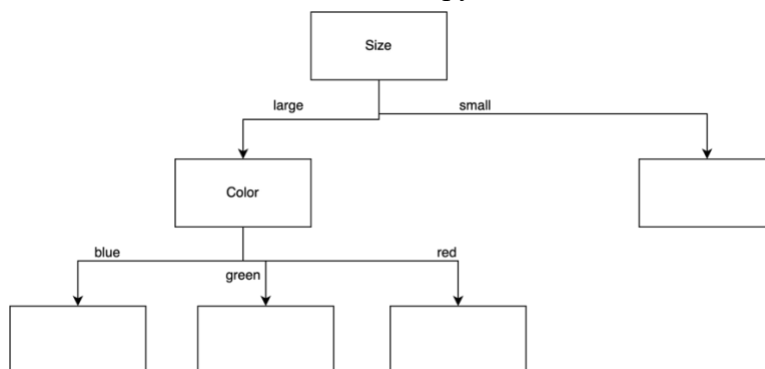
Shape	Size	Color	Safe?
round	large	blue	no
square	large	green	yes
round	large	red	yes
round	large	green	yes
square	large	green	yes

square	large	red	no
square	large	green	yes
round	large	red	yes

we re-evaluate the entropy of each feature.

Feature	Attribute	Label	Label Size	P	Entropy Formula	Entropy Value	Feature Entropy
Shape	round	yes	3	3/4	$-3/4 * \log_2(3/4)$	0.311	$4/8 * 0.311 = 0.156$
Shape	round	no	1	1/4	$-1/4 * \log_2(1/4)$	0.500	$4/8 * 0.500 = 0.250$
Shape	square	yes	3	3/4	$-3/4 * \log_2(3/4)$	0.311	$4/8 * 0.311 = 0.156$
Shape	square	no	1	1/4	$-1/4 * \log_2(1/4)$	0.500	$4/8 * 0.500 = 0.250$
Shape							0.811
Color	blue	yes	0	0/1	0	0.000	$1/8 * 0.000 = 0.000$
Color	blue	no	1	1/1	$-1/1 * \log_2(1/1)$	0.000	$1/8 * 0.000 = 0.000$
Color	green	yes	4	4/4	$-4/4 * \log_2(4/4)$	0.000	$4/8 * 0.000 = 0.000$
Color	green	no	0	0/4	0	0.000	$4/8 * 0.000 = 0.000$
Color	red	yes	2	2/3	$-2/3 * \log_2(2/3)$	0.390	$3/8 * 0.390 = 0.146$
Color	red	no	1	1/3	$-1/3 * \log_2(1/3)$	0.528	$3/8 * 0.528 = 0.198$
Color							0.344

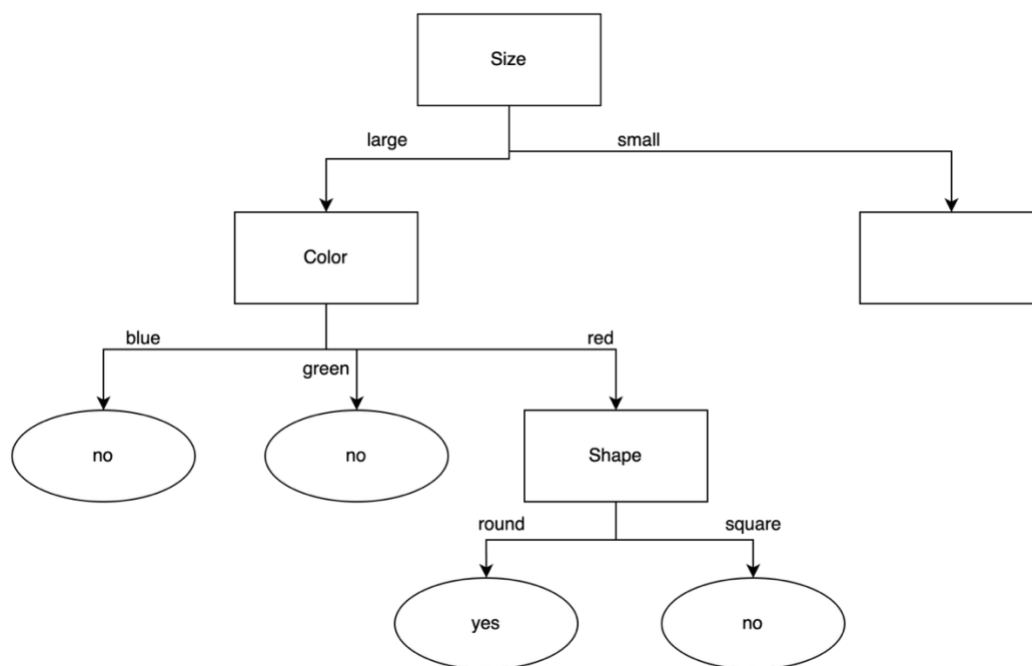
From these values the lowest entropy feature is Color, and we can add that to the tree:



Starting from the left, we remove all non-blue rows which results in a data set with all labels being “no”, so we hit a base case and make the leftmost node a leaf node with the value “no”. The same is true for green as well. For red, we do not hit a base case, so we continue partitioning on the Shape. Since Shape is the only feature left, we can partition by that, resulting in the subset data:

Shape	Size	Color	Safe?
round	large	red	yes
square	large	red	no
round	large	red	yes

We see that all rounds are safe and the square is not, so we can fill in the tree.



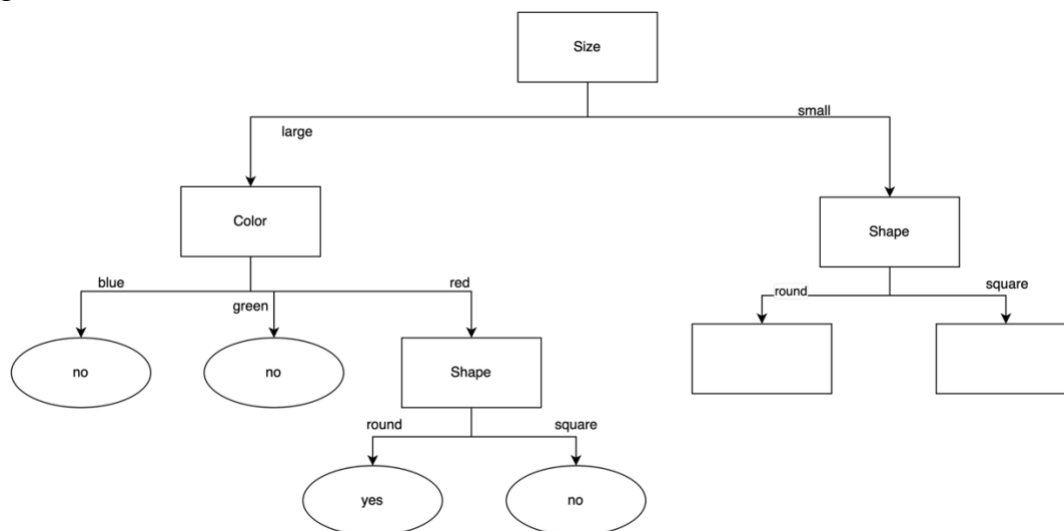
Now, we move back and approach the right child of size (small). Removing the large rows we have data subset:

Shape	Size	Color	Safe?
square	small	red	no
square	small	blue	no
round	small	blue	no
round	small	red	yes
square	small	green	no
square	small	red	no
round	small	green	no

Filling in the entropy table:

Feature	Attribute	Label	Label Size	P	Entropy Formula	Entropy Value	Feature Entropy
Shape	round	yes	1	1/3	$-1/3 * \log_2(1/3)$	0.528	$3/7 * 0.528 = 0.226$
Shape	round	no	2	2/3	$-2/3 * \log_2(2/3)$	0.390	$3/7 * 0.390 = 0.167$
Shape	square	yes	0	0/4	0	0.000	$4/7 * 0.000 = 0.000$
Shape	square	no	4	4/4	$-4/4 * \log_2(4/4)$	0.000	$4/7 * 0.000 = 0.000$
Shape							0.394
Color	blue	yes	0	0/2	0	0.000	$2/7 * 0.000 = 0.000$
Color	blue	no	2	2/2	$-2/2 * \log_2(2/2)$	0.000	$2/7 * 0.000 = 0.000$
Color	green	yes	0	0/2	0	0.000	$2/7 * 0.000 = 0.000$
Color	green	no	2	2/2	$-2/2 * \log_2(2/2)$	0.000	$2/7 * 0.000 = 0.000$
Color	red	yes	1	1/3	$-1/3 * \log_2(1/3)$	0.528	$3/7 * 0.528 = 0.226$
Color	red	no	2	2/3	$-2/3 * \log_2(2/3)$	0.390	$3/7 * 0.390 = 0.167$
Color							0.394

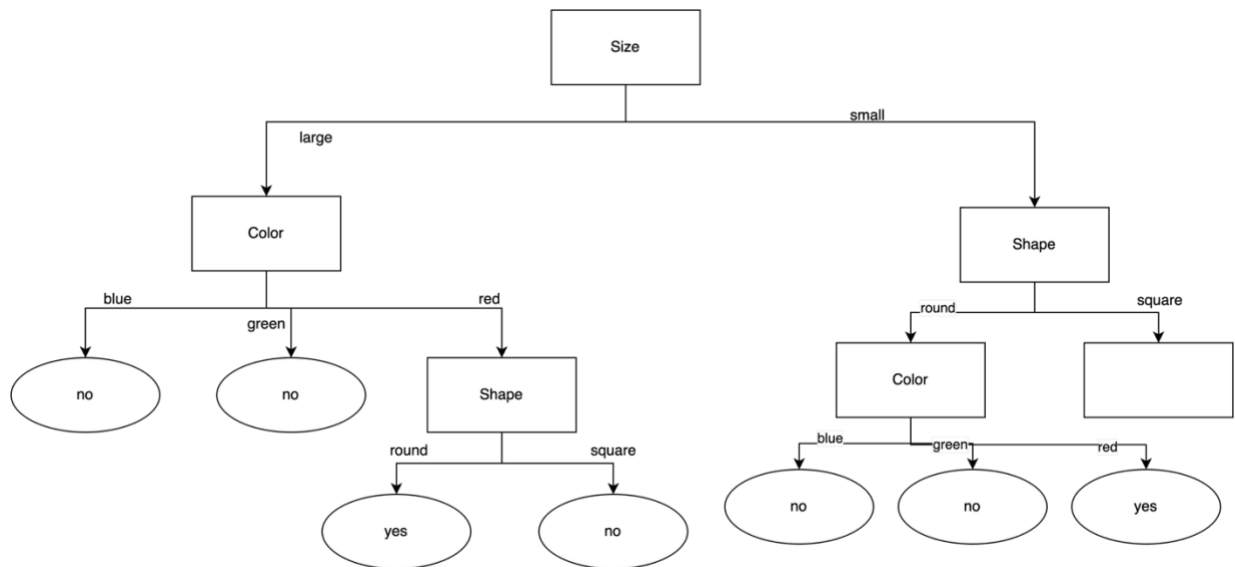
We see that the entropies are the same, so we select shape as the feature to partition on. This gives us the tree:



Partitioning on round we get the data:

Shape	Size	Color	Safe?
round	small	blue	no
round	small	red	yes
round	small	green	no

We can then partition on color and see that blue and green are not safe labels and red is. This results in the tree:



Looking at square, we get the data set:

Shape	Size	Color	Safe?
square	small	red	no
square	small	blue	no
square	small	green	no
square	small	red	no

Since this is a homogenous label set, we mark the node as not safe:

